SAMPLE

Famous Astronomers: Williamina Fleming

The Star Classifier

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Introduction

The most quoted sentence in the history of the Harvard College Observatory was spoken, or is said to have been spoken, in a moment of exasperation, and it concerns the subject of this book directly. Edward Charles Pickering, the observatory's director, frustrated with the work of a male assistant, declared that his maid could do a better job of classifying stellar spectra than the man he was paying to do it. Whether the sentence was spoken exactly that way, and whether the maid overheard it or was told about it later, is a detail the record has not settled.

What the record does settle is everything that followed: the maid was

Williamina Fleming, a twenty-two year old Scottish immigrant, abandoned by her husband in Boston and newly pregnant, working in the director's house, and within a decade of the remark, if it was ever made, she was the director's most important scientific collaborator, the author of a catalogue of more than ten thousand stars, and the supervisor of the team of women who trained a century of astronomy's eyes. The sentence survives because it is the shortest way to tell the whole story, and the story is the subject of this volume of the Famous Astronomers series: a woman who entered astronomy through the servants' entrance of a Boston townhouse and left it through the front door of the profession, carrying with her the star catalogue that made modern classification possible.

The facts of the life are these, and they are evergreen. Williamina Paton Stevens was born in Dundee, Scotland, in 1857, taught school as a teenager, married a widower named James Orr Fleming in 1877, and crossed the Atlantic with him in 1878 while pregnant. The marriage collapsed on arrival: the husband disappeared from the record in Boston, and the abandoned wife, before her twenty-first birthday, was supporting herself as a domestic servant in the house of the Harvard director. Pickering hired her, first for household work and then, discovering in her the precision the remark attributes to her, for the observatory itself.

By 1881 she was on the permanent staff, by 1886 she was the central figure of the Henry Draper Memorial, the privately funded project to photograph and classify the spectra of the northern stars, and in 1890 she published the Draper Catalogue of Stellar Spectra, a compilation of 10,351 stars that established the template every later classification system followed. In 1899 Harvard created for her the Curatorship of Astronomical Photographs, the first corporate appointment the university had ever conferred on a woman, and in 1906 the Royal Astronomical Society elected her an honorary member, one of the first women it had ever so honored.

She died in Boston in 1911, at fifty-four, still in office, having catalogued the spectra of thousands of stars, discovered ten novae, recorded dozens of gaseous nebulae, found hundreds of variable stars, and trained the two women, Annie Jump Cannon and Henrietta Swan Leavitt, whose names stand beside hers in the history of the science.

This book follows the series pattern: the discovery of the woman and her work, the understanding of what she did and how she did it, her key characteristics, the science behind the work, the observation and measurement of her discoveries, the features of the career, her context among the astronomers of her time, the modern understanding of her legacy, the research and exploration she inspired, the ledger of her significant discoveries, the

challenges and mysteries of her story, the future of the questions she opened, and the trivia and significance the record preserves. The chapters keep a fixed distinction between the settled facts, which are given plainly, and the disputed ones, of which the famous remark is only the first, which are marked as such.

The private life is treated where the record holds it, and the record holds less of it than the public career, because the subject of a maid-turned-astronomer was not, in her own century, a person whose letters the archivists collected, and the biographer works with what survived: the observatory's plate notebooks, the catalogues, one remarkable journal kept in 1893, and the testimony of the colleagues and students she trained.

The life deserves the telling for a reason beyond its intrinsic drama. The work Fleming did, the classification of stellar spectra, is the foundation of the science's whole modern structure: the spectral sequence that arranges stars by temperature, and the variable star research built on her methods, are living science, and the way they became living science, through the labor of a woman paid as a clerk and titled as a servant, is the clearest single case the history of astronomy offers of how the actual work of science gets done, by whom, and at what price.

The chapters that follow are, in the end, one argument: that the history of astronomy cannot be told without the Harvard computers, and the Harvard computers cannot be told without their chief, and the chief was the Scottish maid. A sidelight on the first Boston years, which the record holds thinly but holds: the abandoned wife's survival in the winter of 1878 was managed by the era's own machinery, the Scottish network of the city, the charitable institutions, and the employer's market for literate servants, and the son's infancy ran through all of it, in boarding rooms, with the mother working, in the arrangement the single mothers of the poor have always run.

The record's specific facts of these months are few, the hiring date, the

address, the wage, and the general fact is enough: the woman the observatory was about to receive had already survived the century's hardest test for a woman alone,

and the survivor who walked into the Pickering house in 1879 was formed, and the formation is the first fact of the astronomer's biography.

A note on the city she landed in, since the port of entry shaped the entry. Boston in 1878 was the astronomy capital of the new world, the place where the scientific culture of New England had put down its instruments, with the Harvard Observatory the elder institution and the amateur culture, the telescope clubs and the popular lectures, thick on the ground, and the

domestic labor market of the city's professional households put the era's literate immigrant women inside the homes of the men who ran that world. The record's plain statement is that the Pickering household, the director's house on Concord Avenue, was one of perhaps a dozen addresses in the country where a maid could end up an astronomer, and the subject of this book had the fortune, and then had the merit, and the biography holds both, because the life needs both, and the chapters that follow show what the merit did with the fortune.

Chapter 1: The Discovery of Williamina Fleming

The subject was born in Dundee on the fifteenth of May, 1857, into a working family of a city that lived by jute and weaving, and the early record is the ordinary record of a clever girl in nineteenth century Scotland. Her father was a carver and gilder, her mother a housekeeper's daughter, and the education she received was the parish schooling that Dundee offered its ambitious poor, which in her case was enough: by fourteen she was teaching, by the standard account a pupil teacher in the local schools, and at twenty, in 1877, she married a widower of her own city, James Orr Fleming, a man of whom the record preserves, by the accident of what followed, almost nothing.

The emigration came a year after the wedding. The couple sailed for Boston in 1878, with the wife pregnant and the husband, by every account the family later gave, unwilling, and the marriage ended in the receiving city: within weeks or months of the landing, James Fleming walked out of the record and never reappears in it. The abandoned wife was twenty years old, in a foreign country, without support, in the last months of a pregnancy. The son she bore, named Edward after his absent father and known all his life as Edward Preston Fleming, was born in Boston in 1878, and the mother's next documented position is the one the whole story turns on: she was hired, in 1879, into the

household of Edward Charles Pickering, the Harvard College Observatory's director, as a maid.

The household she entered was not an ordinary one. Pickering was forty-three, the head of an observatory he had rebuilt into the world's leading institution for the new science of astronomical photography, and the house on Concord Avenue adjoined the observatory's work in the way that mattered: the director's domestic staff and his scientific staff were, in this period, beginning to be the same labor pool, because the new photographic astronomy required the plate examinations and cataloguing that the era called clerical work and that the era, without embarrassment, assigned to women. The story of Fleming's transfer from the house to the observatory is told in two versions, and both are given here because neither can be excluded.

In the first, the director, exasperated by a male assistant's careless

classification, made the remark about his maid and then, as an act of pique or of genuine experiment, brought the maid to the plates, and the maid's work was better. In the second, the transfer was gradual and practical: the new servant proved literate, numerate, and unsatisfied with scrubbing, the director was a man who noticed competence, and the noticing promoted her. The versions agree on what matters, that the move happened within two years of the hiring and that it was permanent, and the biography of every man in the observatory who resented it, and there were some, is part of the later chapters.

The first observatory work was clerical in the era's sense and scientific in ours. Fleming was set to copying, checking, and reducing, the arithmetic by which the observatory's observations became published tables, and then to the plate work that the new photography made the heart of the science: the glass photographic plates, exposed by the telescopes and developed in the observatory's darkrooms, were examined, annotated, and indexed, and the examining eye that did it well was worth more than the observing eye that took the picture, a truth the period's hierarchy did not recognize and Fleming's career would make unrecognizable. By 1881 she was a permanent member of the scientific staff, by the middle of the decade she was the observatory's expert on the spectrum plates, and by 1886, when the Henry Draper Memorial began, she was the only person in the building, director included, whose command of the full plate collection was complete.

The discovery of Williamina Fleming, as the nineteenth century recorded it, is therefore the discovery of a working method before it is the discovery of a person. The Boston papers that mentioned her, the observatory's own published volumes that she edited, and the directory of the profession that

slowly learned her name all describe the same career in the terms the era had for it: a computer, a clerk, a copyist, a curator. The modern record, which this book uses, describes what the era's terms concealed: the profession's leading

taxonomist, the author of its

standard catalogue, and the manager of its largest scientific workforce. The chapters that follow move between those two descriptions deliberately, because the distance between them is the discovery of the woman, and the biography's task is to close it.

A sidelight on the understanding: the transfer from house to observatory had a middle stage the record keeps, and it calibrates the whole story's economics. The wages of the household position and the wages of the plate room were, in the earliest years, nearly the same, and the change from maid to computer was, in the ledger, barely a change at all: the difference was the work, and the

work's difference is what the biography measures, since the plate room's labor produced the science the world cited and the household's produced the clean house, and the century paid them alike, and the century's blindness on this point is the fact the modern reader should keep through every chapter that follows.

A further element belongs in any honest portrait of her: the class consciousness of the room, which the record preserves and the modern reader needs, because the subject and her colleagues knew exactly what they were. The women of the plate room, in their own writings, describe themselves as workers, not pioneers: the journal's famous accounting is a worker's accounting, of wages, hours, output, and standing, and the room's letters use the era's labor vocabulary without embarrassment, the language of tasks, pay, and the boss. The understanding that the modern reader should take is the one the room had: the computers did not experience the work as an ascent toward glory, they experienced it as a job done extraordinarily well under ordinary conditions, and the glory, which the later century awarded, should be read back onto the job and not the reverse, and the biography's chapters have been written from the job's side.

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